Early Automated Classification System for Skin Cancer Diagnosis using CNN Architectures

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***Abstract*—**Skin cancer can be cured more easily if it is found early. Dermoscopic images help doctors, but manual checking of many images is slow and depends on experience. In this paper, we build a computer-based system that automatically classifies dermoscopic images from the HAM10000 dataset which is separated into 7 types of skin lesions. We use popular deep learning models such as ResNet-18, ResNet-50, and EfficientNet-b0 with transfer learning and replace their last layer to train for this specific task. On top of these models, we add a small channel-attention block that looks at the final feature maps, learn which channels are more important, and reweights them before classifications. This block is light in terms of parameters but improves both accuracy and macro-F1 score, especially for classes that have fewer samples. We also plot attention weights to show which feature channels the model focuses on for a given image, which increases understandability. The complete system is implemented in MATLAB, along with a simple graphical user interface that allows users to load a dermoscopic image, get predicted class and confidence.

***Keywords*—**Skin cancer; dermoscopic images; HAM10000 dataset; deep learning; convolutional neural network (CNN); transfer learning, channel attention, skin lesion classification, MATLAB GUI

I. Introduction

Skin cancer is one of the most dangerous cancers in the world. Among all types, melanoma causes most deaths. If we classify it early, chances of survival will become much higher. But finding cancer from dermoscopic images is not easy. It takes time, and even experienced doctors may give different results. This accuracy is around 60%. Because of this problem, researchers are developing Computer-Aided Systems to help doctors by automatically analyzing skin images [1].

Now-a-days, Deep learning, especially Convolutional Neural Networks (CNN), is giving very good results in medical image analysis. Unlike old methods that use manually designed features, CNN can learn directly from images [14].There are public datasets like HAM10000, which have more than 10,000 skin images in 7 different categories. These datasets help in training and testing models for classification. But still, there are some challenges like unequal data in classes, very small differences between skin diseases, and problems in images like hair and lighting variations [15].

To overcome these challenges, many researchers are using techniques like data augmentation, which helps to increase the number of training images and balance the dataset. Methods such as rotation, flipping, zooming, and brightness adjustment are commonly used [17]. These techniques help the model to learn better and improve its performance on unseen data.

In addition, transfer learning is widely used, where pretrained models like ResNet, and EfficientNet are applied to skin lesion classification [18][13]. These models are already trained on large datasets and can capture important features, which reduce training time and improve accuracy. Fine-tuning these models for medical images gives better results compared to training from scratch.

Some recent approaches also use attention mechanisms to help the model focus more on important regions of the image, such as the lesion area, instead of background noise. This further improves classification performance [16]-[24]. Evaluation of these models is usually done using metrics like accuracy, precision, recall, and F1-score [7][1][2][30][8].

Overall, Deep Learning-based systems show strong potential in assisting dermatologists for early detection of skin cancer [27]. These systems can reduce workload, improve diagnostic accuracy, and provide support in areas where expert doctors are not easily available. In the future, such systems can be integrated into mobile applications and clinical tools for real-time diagnosis and better healthcare support.

II. Related Work

Deep learning is now widely used for automatic analysis of skin images such as dermoscopic and clinical images. Recent research shows a clear shift towards using deeper convolutional neural networks, attention-based techniques, data augmentation methods, and transformer models for skin lesion classification and segmentation. Many studies mainly aim to improve performance on multi-class datasets like ISIC

2019 [3] and HAM10000. Along with accuracy, researchers also focus on handling real-world issues such as imbalance in data classes, noisy inputs, limited dataset size, and the need for better model understanding.

CNN-based classification and transfer learning

A large number of studies use convolutional neural networks along with transfer learning for classifying skin lesions into categories like benign, malignant, or multiple disease types. Popular pre-trained models such as ResNet, DenseNet, and EfficientNet are commonly adapted for this purpose using datasets like ISIC and HAM10000. Usually, the final classification layer of the pre-trained model is modified and replaced with a task-specific fully connected layer followed by a softmax function [1][3][19][29].
These approaches generally achieve good results in both binary and multi-class classification tasks. To further improve performance, researchers apply various image augmentation techniques and sometimes introduce new regularization methods or activation functions. These steps help the model perform better even when the dataset is small or unevenly distributed.

Ensembles and hybrid architectures

Another important research direction involves the use of ensemble and hybrid models to boost classification accuracy. In some works, multiple CNN models [16] are combined, and their outputs are merged using techniques like majority voting or weighted averaging. Such methods often reach or even exceed dermatologist-level performance. In other approaches, CNNs are used together with additional machine learning models such as SVMs or combined with handcrafted feature extraction methods. These create multi-stage or pipeline-based systems. While these methods improve results, they also increase computational complexity and make real-world implementation more challenging.

Attention mechanisms, transformers and metadata fusion

Attention mechanisms have become very common in recent skin lesion classification models. By adding channel and spatial attention modules into CNN architectures like EfficientNet, the model can better focus on important regions and features in the image. This leads to improved accuracy compared to standard CNN models.
Some recent studies use transformer-based models such as Vision Transformers or Swin Transformers. These models are capable of capturing global relationships within images and are sometimes combined with CNNs for better performance [29].
Another advanced approach is metadata fusion, where image features are combined with additional information like patient data or clinical details. This helps improve predicted accuracy. However, these models are usually more complex and require higher computational resources and training time.

III. Proposed Methodology

A. Splitting of HAM 10000 Data Set

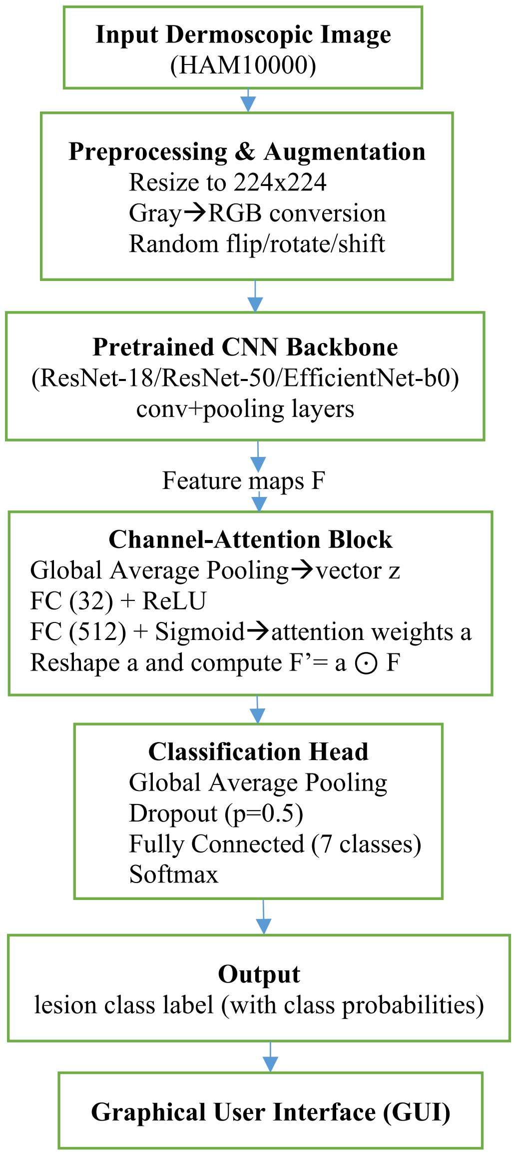
The performance of the proposed methodology was measured on the dermoscopic images which are available on kaggle to researchers, students and experts of HAM10000 dataset. HAM10000 consists of 10,015 images distributed into 7 types of skin lesions.

TABLE-I: TYPES OF SKIN LESIONS AND ITS'S COUNT IN HAM10000 DATASET

Diagnostic Class	Abbreviation	Number of Images
Melanoma	mel	1,113
Melanocytic Nevi	nv	6,705
Basal Cell Carcinoma	bcc	514
Actinic Keratoses & Intraepithelial Carcinoma (or) Bowen's Disease	akiec	327
Benign Keratosis	bkl	1,099
Dermatofibroma	df	115
Vascular Lesions	vasc	142
Total		**10,015**

Data set is divided into 80:20, i.e.,80% fot training and validation and 20% for testing purpose.

B. Process of Proposed Model Algorithm



C. *Performance Parameters*

Accuracy:

It measures how many total predictions are correct out of all samples.

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Precision:

It measures how many predicted positive cases are actually correct.

$$\text{Precision} = \frac{TP}{TP+FP}$$

Recall:

It measures how many actual positive cases are correctly identified.

$$\text{Recall} = \frac{TP}{TP+FN}$$

F1-score:

It gives a balance between recall and precision.

$$\text{F1-score} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Where:

- **TP (True Positive):** Correctly predicted positive cases
- **TN (True Negative):** Correctly predicted negative cases
- **FP (False Positive):** Wrongly predicted as positive
- **FN (False Negative):** Wrongly predicted as negative

D. *Quantitative Results*

We proposed that ResNet-18 and EfficientNet-B0 integrated with an attention mechanism demonstrate strong classification performance on the validation dataset. The model achieves an overall accuracy of approximately ~80-89%, indicating high reliability in distinguishing between different types of skin lesions. Another model, the ResNet-50, is trained without an attention mechanism.

TABLE-II: COMPARATIVE PERFORMANCE FOR DIFFERENT CNN ARCHITECTURES

Pre-trained CNN	No. of epochs	No. of minutes trained	Validation Accuracy	Overall Accuracy
ResNet-18	25	67.08	88.07%	88.02%
ResNet-50	25	202.56	86.97%	86.97%
EfficientNet-b0	25	293.47	87.07%	87.01%
ResNet-18 with Attention Mechanism	30	124.77	87.02%	86.92%
EfficientNet-b0 with Attention Mechanism	75	957.24	88.68%	88.69%

E. *Confusion Matrix Analysis:*

The confusion matrix provides a detailed visualization of classification performance across all classes.

Key observations include:

- Most predictions lie along the diagonal indicating correct classifications.
- Minimal misclassification rates overall.

These misclassifications are expected due to overlapping visual features such as color variation and texture.

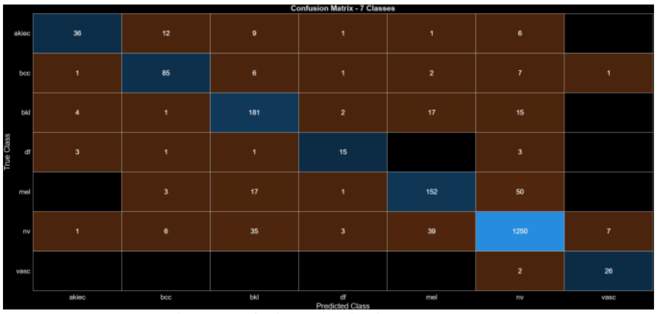


Fig.1 Confusion Matrix for ResNet-18

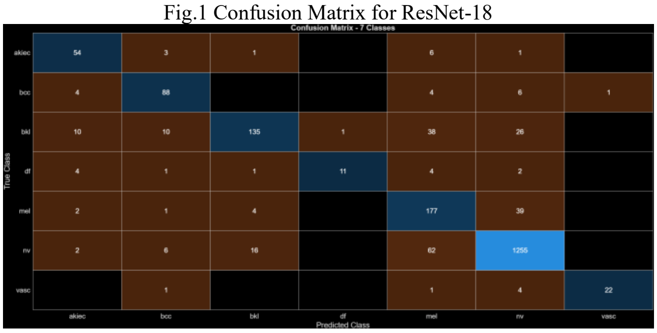


Fig.2 Confusion Matrix for ResNet-50

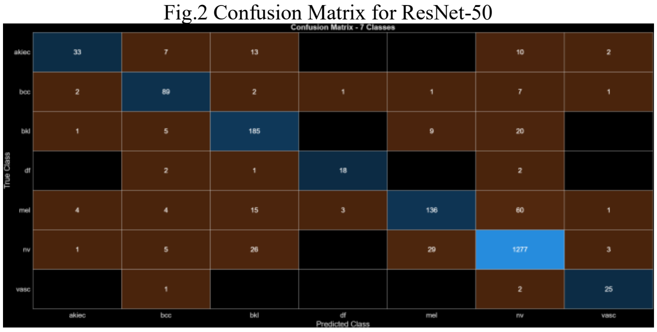


Fig.3 Confusion Matrix for EfficientNet-b0

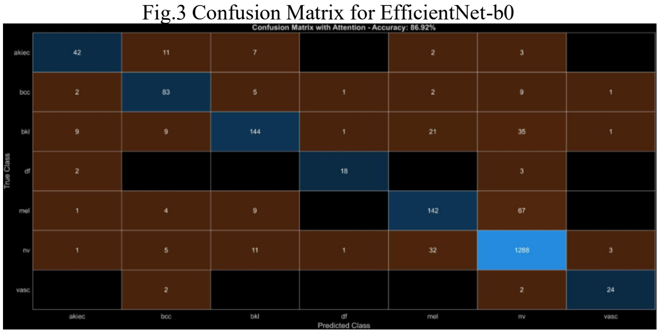


Fig.4 Confusion Matix for ResNet-18 with Attention Mechanism

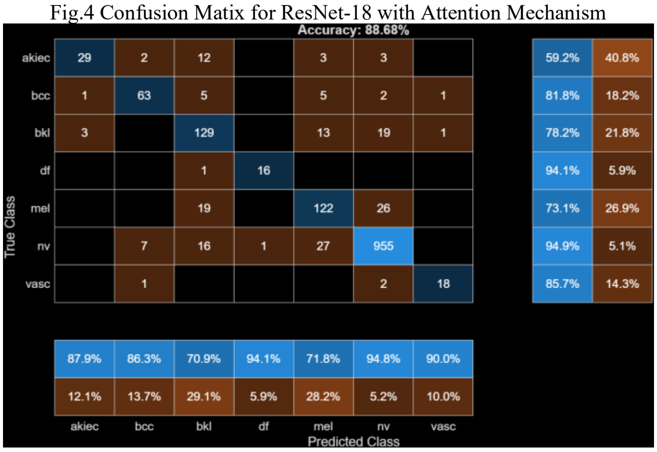


Fig.5 Confusion Matix for EfficientNet-b0 with Attention Mechanism

IV . Experiments & Results

TABLE-III Pre-class performance metrics for different CNN architectures

Pre-trained CNN	Type of Skin Lesion	Recall%	Precision%	F1%
ResNet-18	AKIEC	50.77	80.49	61.7
	BCC	86.41	78.76	82.4
	BKL	84.09	76.45	79.7
	DF	78.26	81.82	70.6
	MEL	60.99	77.71	68.3
	NV	95.23	92.67	93.8
	VASC	89.29	78.12	83.3
ResNet-50	AKIEC	83.08	71.05	76.6
	BCC	85.44	80.00	82.6
	BKL	61.36	85.99	71.6
	DF	47.83	91.67	62.9
	MEL	79.37	60.62	68.9
	NV	93.59	94.15	93.7
	VASC	78.57	95.65	81.5
EfficientNet-B0	AKIEC	55.4	80.0	65.5
	BCC	82.5	78.0	80.2
	BKL	82.3	72.7	77.2
	DF	65.2	65.2	65.2
	MEL	67.9	72.0	69.9
	NV	93.2	93.8	93.5
	VASC	89.7	74.3	81.3
ResNet-18 with Attention Mechanism	AKIEC	64.62	73.68	68.85
	BCC	80.58	72.81	76.50
	BKL	65.45	81.82	72.73
	DF	78.26	85.71	81.82
	MEL	63.68	71.36	67.30
	NV	96.05	91.54	93.74
	VASC	85.71	82.76	84.21
EfficientNet-BO with Attention Mechanism	AKIEC	59.18	87.88	70.73
	BCC	81.82	86.30	84.00
	BKL	78.18	70.88	74.35
	DF	94.12	94.12	94.12
	MEL	73.05	71.76	72.40
	NV	94.93	94.84	94.88
	VASC	85.71	90.00	97.80

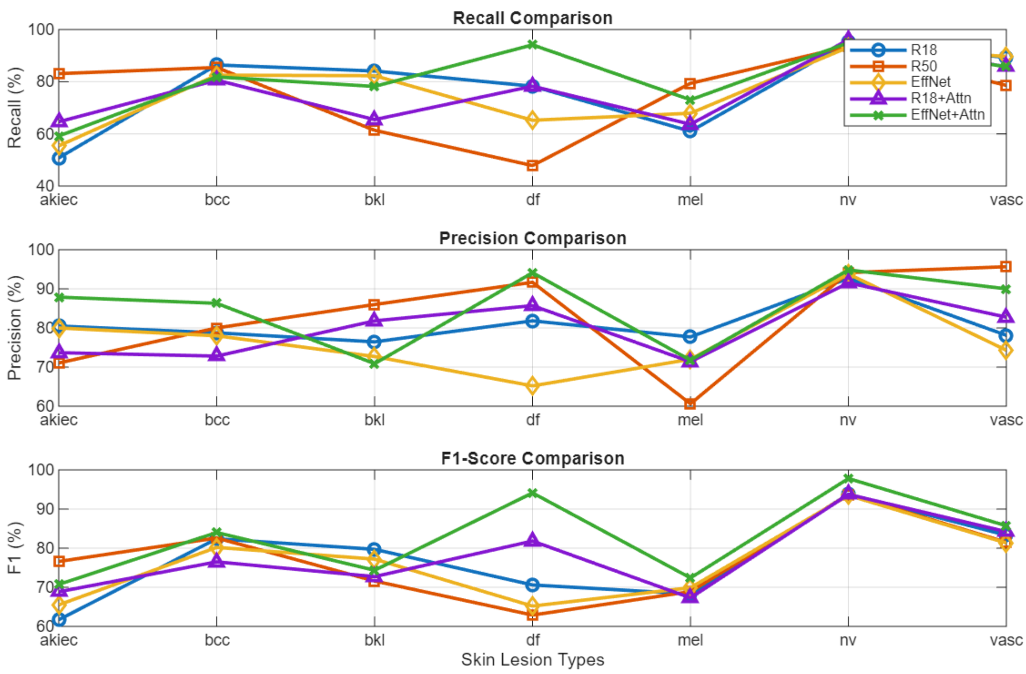


Fig.6 Multi-metric Line Plot for CNN Architectures

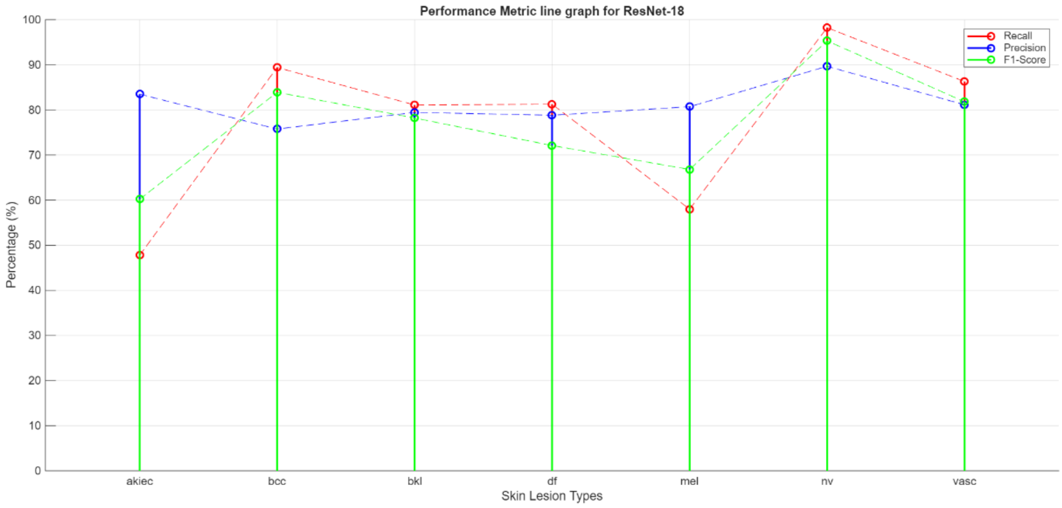


Fig.7

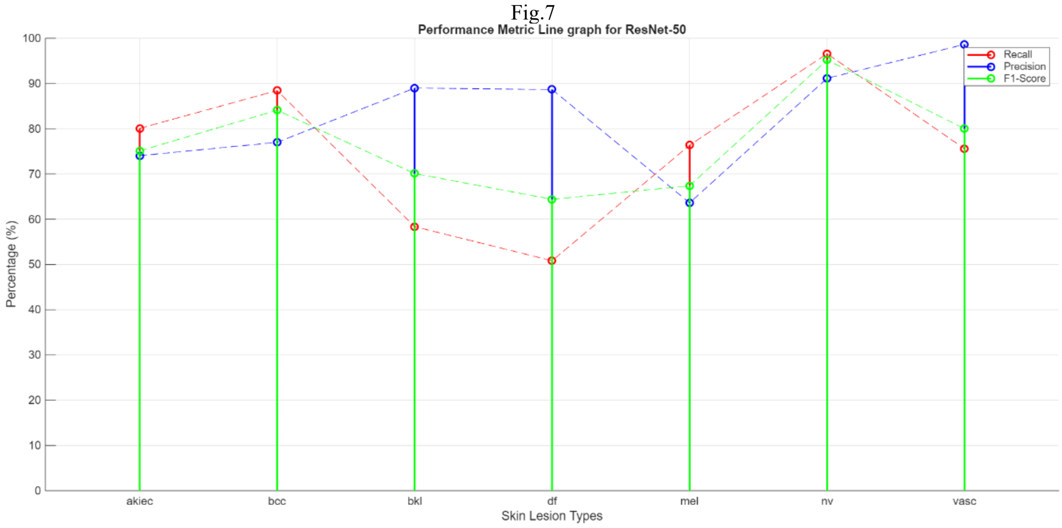


Fig.8

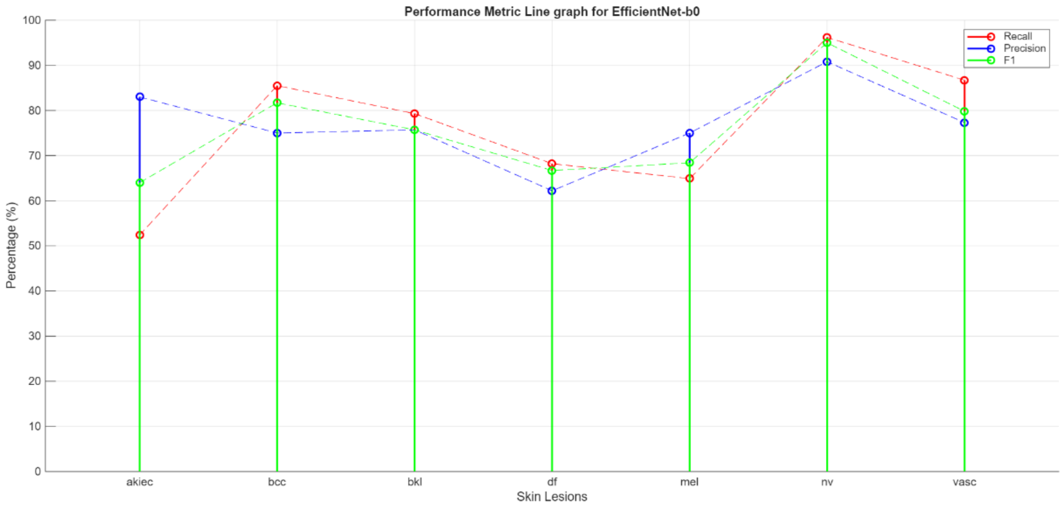


Fig.9

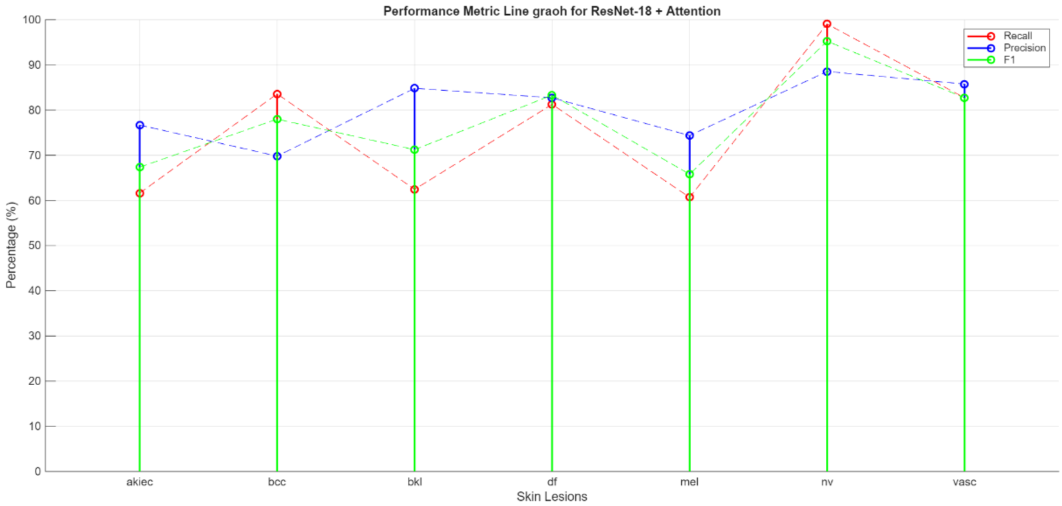


Fig.10

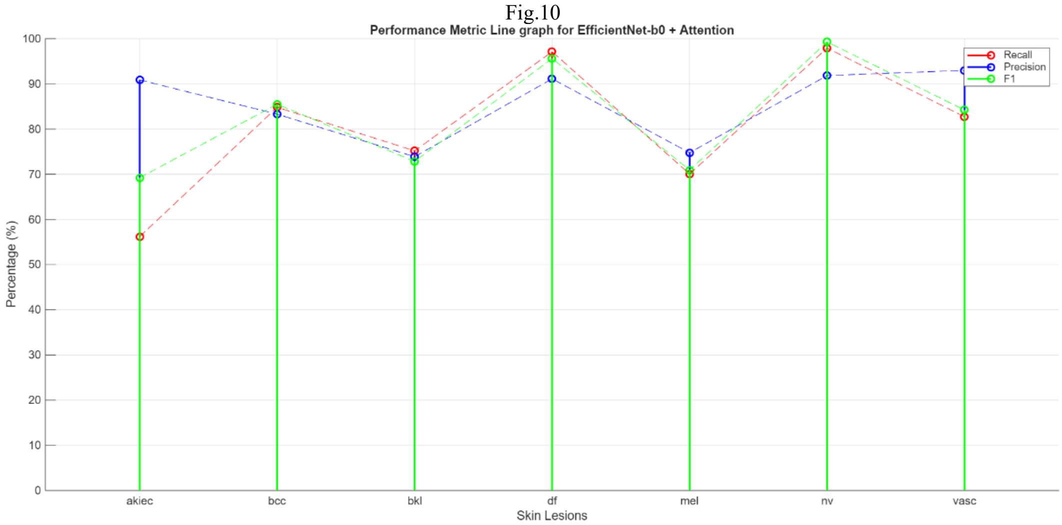


Fig.11

Fig.7-11 Performance Metric Line Graphs for CNN Architectures

Impact of Attention Mechanism:

The integration of the attention module contributes significantly to performance improvement by:

- Enhancing feature representation
- Suppressing irrelevant background information
- Improving classification accuracy, particularly for complex lesions

Experimental results indicate that the attention-augmented model consistently outperforms the baseline ResNet-18 without attention.

Overall, the proposed model demonstrates robust and reliable performance on the HAM10000 dataset. The key outcomes are:

- High validation accuracy (~89%)
- Strong generalization capability
- Effective handling of complex lesion patterns
- Improved feature interpretability through attention visualization

These results confirm that the proposed approach is suitable for automated skin lesion classification, with potential applications in early melanoma detection and clinical decision support systems.

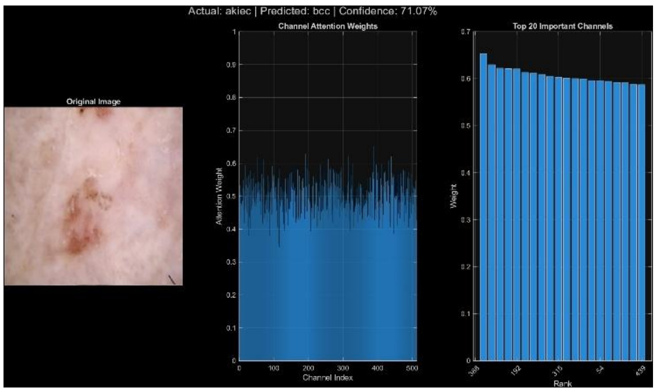


Fig.12 Channel Attention weights for ResNet-18 after model training

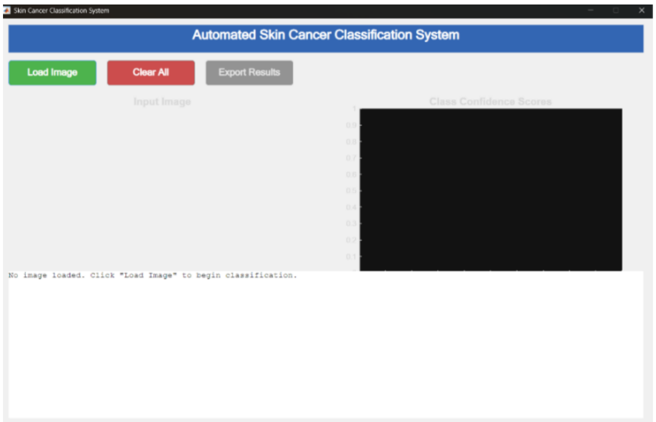


Fig.13 Graphical User Interface before image upload

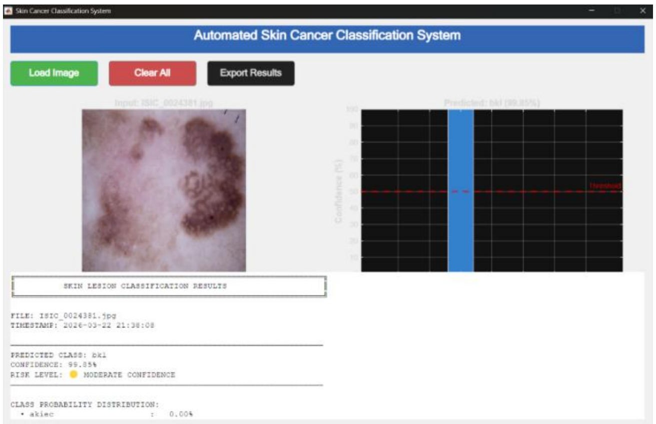


Fig.14 Graphical User Interface after image upload which classifies the type of skin lesion

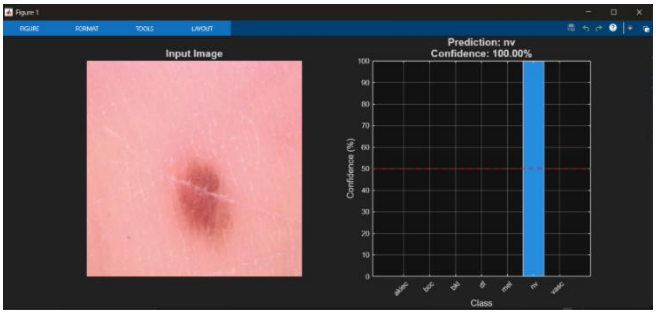


Fig.15 Single Image Prediction

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